

JEE Advanced Booster Test - 1 | 2022

Date: 12/10/2020

Maximum Marks: 180

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

- The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **four** sections (**Section 1, Section 2, Section 3 and Section 4**).
- Section 1** contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

Section 2 contains **2 Paragraphs containing 2 Questions** on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Section 3 contains **2 Column-Match sets**. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Section 4 contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION - 1

- This section consists of **SIX (06)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+4 If only (all) the correct option(s) is(are) chosen
Partial Marks:	+3 If all the four options are correct but ONLY three options are chosen
Partial Marks:	+2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks:	+1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION - 2

- This section has **TWO (02) Paragraphs containing TWO (02) Questions** on each paragraph. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct option is chosen
Zero Marks:	0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION - 3

- This section has **TWO (02) Column-Match sets**. Each set contains statements given in two columns. Statements in column I have to be matched with statements in column II. One or more than one statements in column II can be matched with a statement in column I.
- Each set has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the option corresponding to the correct combination is chosen
Zero Marks:	0 If none of the options is chosen (i.e., the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION 4

- This section contains **6 Integer Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

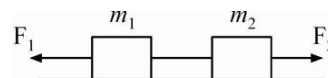
Full Marks:	+3 If ONLY the correct Integer value is entered. There is NO negative marking .
Zero Marks:	0 In all other cases.

PART I : PHYSICS**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

1. Two blocks connected by a massless rope are kept on a frictionless floor. Now opposing forces F_1 and F_2 are applied on the two blocks as shown. Let tension in the connecting rope be T . Choose correct options.

- (A) If $m_1 = m_2$, T is equal to $|F_1 - F_2|$
 (B) If m_1 is much greater than m_2 , T is nearly equal to F_2
 (C) If m_2 is much greater than m_1 , T is nearly equal to F_2
 (D) If $m_1 = m_2$, T is equal to $\frac{F_1 + F_2}{2}$

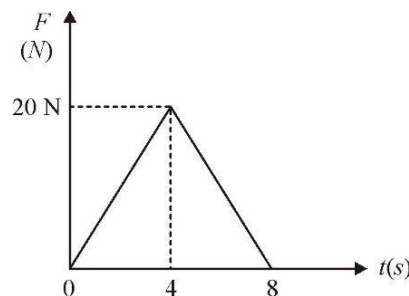


2. A set of n identical cubical blocks lie at rest parallel to each other along a line on a smooth horizontal surface. The separation between the near surface of any two adjacent blocks is L . The block at one end is given a speed V towards the next one at $t = 0$. Choose the correct statements.

- (A) If all the collisions are completely inelastic, then the last block starts moving at $t = n(n-1) \frac{L}{2V}$
 (B) If all the collisions are elastic, then the last block starts moving at $t = (n-1) \frac{L}{V}$
 (C) The centre of mass of the system will have a final speed $\frac{V}{n}$, if all the collisions are completely inelastic
 (D) The centre of mass of the system will have a final speed V , if all the collisions are elastic

3. A box of mass 4kg placed on a rough horizontal floor, is acted upon by a horizontal force F that varies with time as shown in the given graph. If coefficients of static and kinetic frictions are 0.25, on which of following conclusions can you arrive at? (Take $g = 10 \text{ ms}^{-2}$)

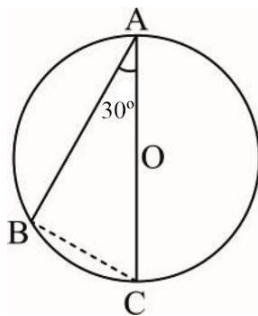
- (A) The box starts moving at the instant $t = 2.0 \text{ s}$
 (B) The maximum velocity acquired by the box is 5.0 m/s
 (C) The box stops at the instant $t = 9.0 \text{ s}$
 (D) The maximum velocity acquired by the box is 2.5 m/s



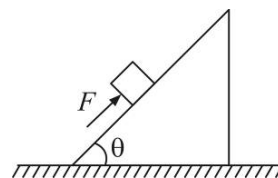
4. A particle of unit mass is moving along x-axis under the influence of a force and its total energy is conserved. Potential energy function for the particle is given as : $U(x) = \frac{U_0}{2} \left[\frac{x}{a} - \frac{1}{3} \left(\frac{x}{a} \right)^3 \right]$, where U_0 and a are positive constants. Which of the following statements are correct?

- (A) Force acting on the particle is zero at $x = \pm a$
 (B) Force acting on the particle is zero at $x = 0$
 (C) The particle experiences an attractive force towards $x = 0$ in the region $|x| < a$
 (D) $x = -a$ is a position of stable equilibrium

5. A bead is free to slide down on a smooth straight wire AB or AC fixed between two points on a vertical circle. The bead is released from rest from the highest point A of the wire. Choose the correct options.



- (A) Acceleration of the bead is more while sliding down on AC than AB
 (B) Time taken to slide down the wire AB is less than that for AC
 (C) Time taken to slide down the wire AB is equal to that for AC
 (D) The speed with which the bead reaches the other end of wire is more for AC than AB
6. A force F is applied on a block of mass m kept on a rough, fixed inclined plane. The direction of F is parallel to the inclined surface. The coefficient of static friction is μ such that $\mu < \tan \theta$. Choose the correct statements.
- (A) The block slides down the plane if $F < mg \sin \theta$
 (B) The block slides up the plane if $F > mg \sin \theta + \mu mg \cos \theta$
 (C) The block slides down the plane if $F = 0$
 (D) No friction acts on the block if $F = mg \sin \theta$



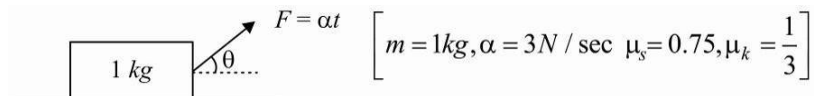
SPACE FOR ROUGH WORK

SECTION 2

This section has TWO (02) Paragraphs containing TWO (02) Questions on each paragraph. Each question has FOUR options. ONLY ONE of these four options is the correct answer.

Paragraph for Q.7 to Q.8

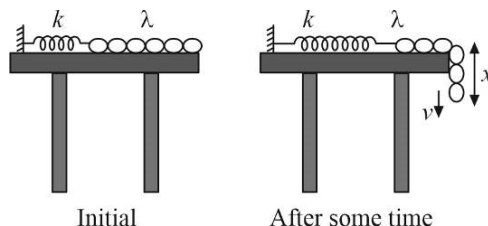
A variable force $F = \alpha t$ is acting on block of mass 'm' at an angle θ with horizontal as shown.



7. Minimum possible time after which block can start moving is:
 (A) 1sec (B) 2sec (C) 3sec (D) Data insufficient
8. Speed of block when it leaves the horizontal surface for $\theta = 37^\circ$ is approximately:
 (A) 20 m/s (B) 25 m/s (C) 28 m/s (D) 26 m/s

Paragraph for Q.9 to Q.10

A chain has mass per unit length $\lambda = 0.1 \text{ kg/cm}$ and total length 1 m. It is kept on a smooth table and its one end is connected to a spring of spring constant $k = 50 \text{ N/m}$. The spring is in relaxed state and its other end is connected to a support fixed on the table. The chain is disturbed slightly so that it starts falling from the table.



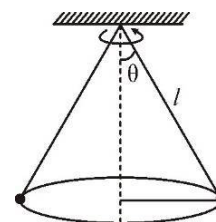
9. As the chain falls down, its acceleration _____. (before it slides off from the table completely)
 (A) increases (B) decreases
 (C) remains constant (D) increases then decreases
10. The speed of the hanging part as a function of its length x is given by: (v and x are in SI unit)
 (A) $v = 5x$ (B) $v = \sqrt{5}x$ (C) $v = 5x^2$ (D) $v = \sqrt{10}x$

SPACE FOR ROUGH WORK

SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. A ball of mass m is connected to a string of length l . If the ball is rotating in a horizontal circle such that string is making angle ' θ ' with vertical. Then match the unknowns in Column-I with their corresponding values in Column-II.



$$(\theta = 37^\circ, l = 0.5m, g = 10 \text{ m/s}^2, m = 2 \text{ kg})$$

	Column 1		Column 2
(P)	Speed of ball is (m/sec)	1.	5
(Q)	Tension in string (N)	2.	$\frac{3}{2}$
(R)	Acceleration of ball (ms^{-2})	3.	25
(S)	Angular speed of ball. (rad s^{-1})	4.	$\frac{15}{2}$

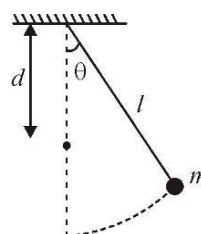
(A) P - 2 ; Q - 3 ; R - 4 ; S - 3

(B) P - 2 ; Q - 3 ; R - 4 ; S - 1

(C) P - 1 ; Q - 4 ; R - 4 ; S - 1

(D) P - 2 ; Q - 3 ; R - 2 ; S - 1

12. In the given system, when the ball of mass m is released, it will swing down the dotted arc. A nail is located at a distance d vertically below the point of suspension.



(Take $g = 10 \text{ m/s}^2$)

	Column I		Column II
(P)	$\theta = 60^\circ, d = 0.6l$	1.	Ball will swing completely around the circle centered along the nail
(Q)	$\theta = 53^\circ, d = 0.6l$	2.	Ball will not be able to swing the complete circle centered along the nail.
(R)	$\theta = 60^\circ, d = \frac{41}{50}l$	3.	Tension in string just after it touches the nail is $3.5mg$
(S)	$\theta = 53^\circ, d = \frac{41}{50}l$	4.	Tension in string just after it touches the nail is $3mg$

(A) P - 1, 3 ; Q - 2, 4 ; R - 1 ; S - 2

(B) P - 2 ; Q - 2 ; R - 1 ; S - 1

(C) P - 2, 3 ; Q - 2, 4 ; R - 1 ; S - 2

(D) P - 2, 3 ; Q - 2, 4 ; R - 2 ; S - 2

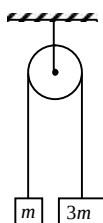
SPACE FOR ROUGH WORK

SECTION 4

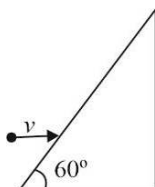
This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.

(Example: 6, 81, 1.50, 3.25, 0.08)

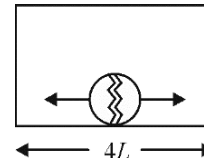
13. In the figure shown, pulley and string are massless. The blocks move in vertical plane due to gravity. Find the magnitude of acceleration of centre of mass of the blocks (in m/s^2). (Take $g = 10 m/s^2$)



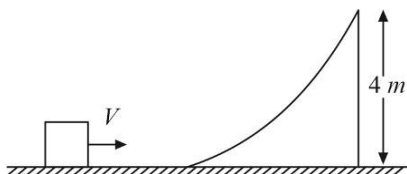
14. As shown in figure, a small ball strikes a smooth fixed incline horizontally. After collision, its velocity is directed vertically upwards. Find the coefficient of restitution for the collision.



15. A bomb of mass $3 kg$ is kept inside a closed box of mass $3 kg$ and length $60 cm$ at its centre. It explodes in two parts of mass $1 kg$ and $2 kg$. The two parts move in opposite direction along the surface of box and stick to the opposite side of the walls of box. Box is kept on a smooth horizontal surface. What is the distance (in cm) moved by the box during this time interval?



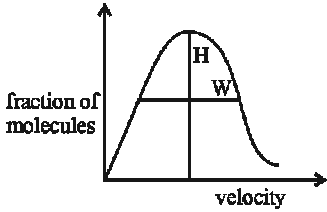
16. A force $\vec{F} = 3xy^2 \hat{i} + y \hat{j}$ is acting on a particle which moves along the curve $y = x^2$. Find work done by $\vec{F}$ during the movement of particle from $x = 0$ to $x = 2$. All given quantities are in SI units and answer is also to be reported in SI unit.
17. A particle of mass $0.2 kg$ is moving in one dimension under a force that delivers a constant power $0.5 W$ to the particle. If the initial speed of the particle is zero, the speed (in ms^{-1}) after $5s$ is _____.
18. A small block of mass $1 kg$ is projected towards a smooth movable wedge of mass $4 kg$. The horizontal surface is also smooth. Find the minimum speed of projection of block V (in m/s) so that the block is able to reach the top of wedge which is at a height of $4 m$ above the surface. (Take $g = 10 m/s^2$)



SPACE FOR ROUGH WORK

PART II : CHEMISTRY**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- Which of the following is(are) correct statements(s) ?
 - In NO_4^+ , all four O atoms are bonded through single bonds with N
 - For square planar molecular geometry, hybridization state of central atom is dsp^2
 - Molecular shape of XeF_5^- is pentagonal planar
 - Hybridization state of central atom in SNF_3 is sp^3
- At the end point in acid-base titrations of a basic substance with acid using phenolphthalein pink-violet colour of the solution disappears, while using Methyl orange, a red colour appears. During the titration of a mixture of Na_2CO_3 and NaHCO_3 with HCl using phenolphthalein and methyl orange indicators one after the other respectively:
 - disappearance of pink-violet colour corresponds to the neutralization of NaHCO_3
 - disappearance of pink-violet colour corresponds to the conversion of Na_2CO_3 to NaHCO_3
 - appearance of red colour corresponds only to the complete decomposition of NaHCO_3 previously present
 - appearance of red colour corresponds to the decomposition of NaHCO_3 formed and that previously present
- Tetrahedral geometry is appropriate for the underlined atom in :
 - I_2Cl_6
 - $\text{N}(\text{SiH}_3)_3$
 - $(\text{CH}_3)_4\text{N}^+$
 - $(\text{CH}_3)_2\text{SO}$
- Molecular chlorine, Cl_2 , has a smaller ionization energy than atomic chlorine, Cl ($\text{IE} = 1108 \text{ kJ mol}^{-1}$ for Cl_2 and 1256 kJ mol^{-1} for Cl). Which is/are the best explanation for this difference?
 - Cl has a higher electron affinity while Cl_2 does not.
 - Cl_2 is diamagnetic while Cl is paramagnetic.
 - The electron ionized from Cl_2 comes from an orbital that has antibonding character.
 - The electron ionized from Cl_2 comes from an orbital that has higher energy.
- According to molecular orbital theory which of the following transformations is(are) correctly matched with change in property indicated?
 - $\text{C}_2 \rightarrow \text{C}_2^{2-}$; change in number of σ -bonds
 - $\text{O}_2 \rightarrow \text{O}_2^{2-}$; change in magnetic properties
 - $\text{NO} \rightarrow \text{NO}^+$; Stability increases
 - $\text{O}_2 \rightarrow \text{O}_2^-$; Bond length increases
- The Maxwell-Boltzmann distribution for molecular speeds is shown in the following figure. In the figure, **H** is the height of the peak, **L** is the location of the maximum speed, **A** is the area under the curve and **W** is the width at half height. As the temperature is increased, identify the correct statement(s).
 
 - H decreases
 - L increases
 - W increases
 - A remain unchanged

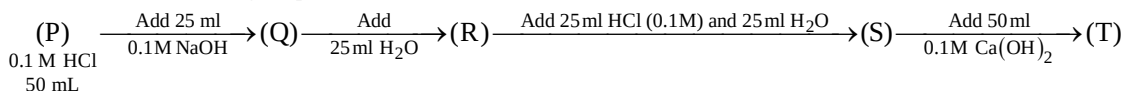
SECTION 2

This section has **TWO (02) Paragraphs** containing **TWO (02) Questions** on each paragraph. Each question has **FOUR options**. **ONLY ONE** of these four options is the correct answer.

Paragraph for Q-7 to Q-8

Monoprotic acids, also known as monobasic acids, are those acids that are able to donate one proton per molecule during the process of dissociation (sometimes called ionization) symbolized by HA eg HCl. Polyprotic acids, also known as polybasic acids, are able to donate more than one proton per acid molecule, in contrast to monoprotic acids that only donate one proton per molecule. Specific types of polyprotic acids have more specific names, such as diprotic (or dibasic) acid (two potential protons to donate), and triprotic (or tribasic) acid (three potential protons to donate). Polyprotic acid is symbolized by H_nX eg H_2SO_4 . Neutralization is the reaction between an acid and a base, producing a salt and neutralized base; for example, hydrochloric acid and sodium hydroxide form sodium chloride and water: When an acid reacts with metal, a salt and hydrogen are produced: acid + metal $\rightarrow$ salt + hydrogen. An example: nitric acid + calcium $\rightarrow$ calcium nitrate + hydrogen. The salt that is produced depends upon which acid and which metal react.

7. Consider the following sequence of information.

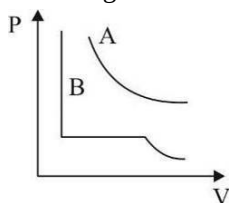


Which of the following are correct?

- (I) Solution (Q) has $[H^+] = 0.03M$
- (II) Solution (Q) & (S) has same concentration of $[H^+]$
- (III) Solution (T) is basic in nature
- (IV) If 40 ml of solution (T) is mixed with 5 ml of (P), then final solution is neutral
- (A) Only (I) (B) (I) (II) and (IV)
- (C) (I), (II) and (III) (D) (I), (II), (III) and (IV)
8. Al is a trivalent metal which reacts with acids to form hydrogen and a salt. 2 mol of Al react completely with a certain acid H_nX ($n = 1, 2$ or 3) to produce 3 mol of hydrogen gas. Which of the following formula(s) of the salt match this question?
- (I) AlX (II) Al_2X_6 (III) Al_2X_3 (IV) AlX_3
- (A) Only (I)
- (B) Only (IV)
- (C) Only (II) and (IV)
- (D) All formula of the salt correctly match with this question

Paragraph for Q-9 to Q-10

For two gases A and B, P v/s V isotherms are drawn at T Kelvin as shown.



9. Which of following is NOT true?
- (A) Ratio of critical temperature of gas B and its Boyle temperature is 8: 27.
 - (B) Gas A behaves as an ideal gas.
 - (C) Gas B behaves as van der Waal's gas.
 - (D) Gas A and B both behaves as ideal gas.
10. The correct statement(s) is/are:
- (I) Pressure correction term will be more negligible for gas B at TK
 - (II) The curve for gas 'B' will be of same shape as for gas A if $T > T_c$ (critical temperature of B)
 - (III) Gas 'A' will show same P v/s V curve as of gas 'B' if $T > T_c$ (critical temperature of A)
 - (IV) Repulsive forces are dominating forces for gas B
- (A) III only (B) II and III (C) II only (D) (I) and (IV)

SPACE FOR ROUGH WORK

SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. MATCH THE FOLLOWING:

	Column 1		Column 2
(I)	C ₂ H ₄	(p)	3c – 2e bonds
(II)	B ₂ H ₆	(q)	Molecular plane is nodal plane for π -bond
(III)	N ₂ H ₄	(r)	Non-polar molecule
(IV)	Al ₂ Cl ₆	(s)	Coordinate bond
		(t)	Non-planar molecule

Choose the correct option:

- (A) [(I)-q] [(II)-p, r, t] [(III)-r, t] [(IV) - s, t]
 (B) [(I)-q, r] [(II)-p, r, t] [(III)- t] [(IV)-r, s, t]
 (C) [(I)-p, q] [(II)-p, r, t] [(III)-q, r, t] [(IV)-p, r, s, t]
 (D) [(I)-q, r] [(II)-p, r, s, t] [(III)-q, r, t] [(IV)-r, s, t]

12. MATCH THE FOLLOWING:

	Column 1		Column 2
(I)	If only force of attraction among the gas molecules be negligible	(p)	$\left(P + \frac{a}{V^2}\right)(V - b) = RT$
(II)	If only the volume of the gas molecules be negligible	(q)	$PV = RT - \frac{a}{V}$
(III)	If both force of attraction and volume of gas molecules is negligible	(r)	$PV = RT + Pb$
(IV)	If both the force of attraction and volume of gas molecule is significant	(s)	$PV = RT$
		(t)	$V_m > 22.4$ (at STP)

Choose the correct option:

- (A) [(I)-r, t] [(II)-q, t] [(III)-s] [(IV)-p, t]
 (B) [(I)-r, t] [(II)-q] [(III)-s] [(IV)-p]
 (C) [(I)-p, t] [(II)-q, t] [(III)-s] [(IV)-p, s, t]
 (D) [(I)-s, t] [(II)-q] [(III)-s] [(IV)-p]

SPACE FOR ROUGH WORK

SECTION 4

This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

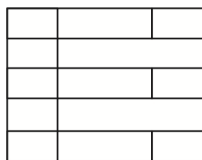
13. According to molecular orbital theory of bonding how many of the following diatomic molecular species have pi bond(s)?
 Be_2^{2-} , B_2 , C_2 , N_2 , O_2 , F_2^{2+} , N_2^+ , CN^- , N_2^{2+} , HF
14. In tetracyanoethylene $\text{C}_2(\text{CN})_4$, number of sp hybridized carbon atoms are X_1 , number of σ bonds are X_2 , number of π bonds are X_3 and number of lone pair of electrons are X_4 . Find out value of $X_1+X_2+X_3+X_4$.
15. If a chemist requires 10 moles of liquid methanol (CH_3OH) for a chemical reaction, what volume (in mL) should be used? The density of CH_3OH is 0.8 g/cm^3 at 25°C .
[Atomic mass of H = 1, C = 12, O = 16 gm/mol]
16. Weight of O_2 necessary to fill up a cylinder of 0.082 litre capacity at 0°C and 96 atm pressure would be____: (Compressibility factor = 0.96 and gas constant (R) = $0.082 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$)
17. Find the vapour density of the gas among the gases (a) to (d) which has the smallest mean free path under the same external conditions? [Atomic mass H =1, N = 14, O =16, Cl = 35.5 g/mol]
(a) H_2 (b) Cl_2 (c) N_2 (d) O_2
18. Two gases A and B with molecular masses M and 2M respectively are mixed in molar ratio 2 : 1. The ratio of rate of effusion of the mixture with respect to the rate of effusion of gas (A) under identical conditions of pressure and temperature will be_____. (Take $\sqrt{3} = 1.732$)

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- If $T_n = \sin^n \theta + \cos^n \theta$, then $\frac{T_6 - T_4}{T_6} = m$ holds for values of m satisfying ($\theta \in R$):
 (A) $m \in \left[-1, \frac{1}{3}\right]$ (B) $m \in \left[0, \frac{1}{3}\right]$ (C) $m \in [-1, 0]$ (D) $m \in \left[-1, -\frac{1}{2}\right]$
- The equation $\sin^4 x + \cos^4 x + \sin 2x + K = 0$ must have real solution if:
 (A) $K = 0$ (B) $|K| \leq \frac{1}{2}$ (C) $-\frac{3}{2} \leq K \leq \frac{1}{2}$ (D) $-\frac{1}{2} \leq K \leq \frac{3}{2}$
- Let N be the maximum number of elements of a subset L of $\{1, 2, 3, \dots, 1999\}$ such that difference of any two distinct elements of L is not equal to 4. The value of N is less than:
 (A) 999 (B) 1002 (C) 1001 (D) 2000
- Number of ways in which $A A B B B C$ can be placed in the squares of the figure as shown, so that no row remains empty, is:



- (A) 81 (B) 4860 (C) $81 \left(\frac{6!}{2!3!} \right)$ (D) 1620
- If $x = \cos 39^\circ + \sin 57^\circ$, $y = \cos 40^\circ + \sin 58^\circ$ and $z = \cos 41^\circ + \sin 59^\circ$, then:
 (A) $x > y$ (B) $y > z$ (C) $x = y = z$ (D) $x + z > y$
- If letters of the word "THING" are arranged in all possible manner and words thus formed are written in dictionary order. If K is the number of words lying between "NIGHT" and "THING" (both exclusive) in that dictionary then:
 (A) Number of zeros at the end of $K!$ is 4
 (B) Number of divisors of K is 6
 (C) Number of integral coordinates (both abscissa and ordinate integer) lying strictly inside triangle formed by $y = 0$, $x = 0$, $x + y = K$ is 171.
 (D) K does not divide number of words in that dictionary

SPACE FOR ROUGH WORK

SECTION 2

This section has TWO (02) Paragraphs containing TWO (02) Questions on each paragraph. Each question has FOUR options. ONLY ONE of these four options is the correct answer.

Paragraph for Q-7 to Q-8

There are two Indian couples, 2 American couples and one unmarried person.

7. The total number of ways in which they can sit around a circular table such that an Indian Husband & an American Husband are on either side of the unmarried person.
 (A) 4320 (B) 20160 (C) 5760 (D) 11520
8. If all the nine persons are to be interviewed one by one, then the total number of ways of arranging their interviews such that no wife gives interview before her husband.
 (A) 24230 (B) 28410 (C) 40320 (D) 22680

Paragraph for Q-9 to Q-10

Consider an equation $\sin x + \sin y = 2$

We know that $\sin x \leq 1$ and $\sin y \leq 1$ for all x, y . So, $\sin x + \sin y \leq 2$ for all x and y i.e., $\sin x + \sin y = 2$ if and

only if $\sin x = 1 = \sin y$ so, $x = 2n\pi + \frac{\pi}{2}$ and $y = 2m\pi + \frac{\pi}{2}$.

In general, one or more of the following extreme value conditions

- I. $-1 \leq \sin x \leq 1 \Rightarrow |\sin x| \leq 1$ and $\sin^2 x \leq 1$
- II. $-1 \leq \cos x \leq 1 \Rightarrow |\cos x| \leq 1$ and $\cos^2 x \leq 1$
- III. $-\sqrt{a^2 + b^2} \leq (a \sin x + b \cos x) \leq \sqrt{a^2 + b^2} \Rightarrow |a \sin x + b \cos x| \leq \sqrt{a^2 + b^2}$

On the basis of above passage, answer the following questions:

9. Number of roots of the equation $\cos^7 x + \sin^4 x = 1$ in the interval $[0, 2\pi]$ is:
 (A) 0 (B) 1 (C) 2 (D) 4
10. The smallest positive number P for which the equation $\cos(P \sin x) = \sin(P \cos x)$ has a solution in $[0, 2\pi]$ is:
 (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4\sqrt{2}}$ (D) $\frac{\pi}{2\sqrt{2}}$

SPACE FOR ROUGH WORK

SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. MATCH THE COLUMN :

Column 1		Column 2	
(A)	The least difference between the roots, in the first quadrant $\left(0 \leq x \leq \frac{\pi}{2}\right)$, of the equation $4 \cos x(2 - 3 \sin^2 x) + (\cos 2x + 1) = 0$ is	(p)	2
(B)	Number of positive values of x for which $2 \cos x, \cos x $ and $1 - 3 \cos^2 x$ are in G.P., lying in $[0, \pi]$ is	(q)	$\frac{\pi}{6}$
(C)	The values of x between 0 and 2π which satisfy the equation $\sin x \sqrt{8 \cos^2 x} = 1$ are in A.P. with common difference d , then $\frac{4d}{\pi}$ is	(r)	$\frac{\pi}{3}$
(D)	The number of solution of the equation $ \sin \theta + \cos \theta ^{(1 + \sin 2\theta)} = 2, -\pi \leq \theta \leq \pi$, is	(s)	1

- (A) [A-r] [B-s] [C-p] [D-s] (B) [A-q] [B-p] [C-s] [D-p]
 (C) [A-q] [B-p] [C-s] [D-s] (D) [A-r] [B-s] [C-p] [D-p]

12. Consider all possible permutations of the letters of the word *INDIANOIL* then :

Column 1		Column 2	
(A)	Number of permutations containing the word INDIA is	(p)	$2 \times 5!$
(B)	Number of permutations in which letter I occurs in first and the last position is	(q)	$3 \times 5!$
(C)	Number of permutations in which none of the letters D, L, N occurs in the last 5 positions is	(r)	$5!$
(D)	Number of permutations in which the letters, A, I, O occurs only in odd position is	(s)	$21 \times 5!$

- (A) [A-r] [B-s] [C-p] [D-q] (B) [A-r] [B-s] [C-p] [D-p]
 (C) [A-p] [B-q] [C-q] [D-p] (D) [A-r] [B-s] [C-q] [D-q]

SPACE FOR ROUGH WORK

SECTION 4

This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

13. If the set of all values of x in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying $|4\sin x + \sqrt{2}| < \sqrt{6}$ is $\left(\frac{a\pi}{24}, \frac{b\pi}{24}\right)$ then value of $\left|\frac{a+b}{3}\right|$ is ____.
14. 17 coupons are numbered from 1 – 17. 6 coupons are selected at random one by one with replacement. If the possible ways such that the largest number appeared on selected coupons is 5 is represented by p . Find the value of $\frac{p}{1500}$.
15. An intelligence agency forms a code of two distinct digits selected from 0, 1, 2, ..., 9 such that the first digit of the code is non-zero. The code hand written on a slip, can however potentially create confusion, when read upside down for example, the code 91 may appear as 16. Let N be the number of codes for which no such confusion can arise. Find sum of digits of N .
16. If maximum value of the expression $\frac{1}{(11\sin^2\theta + 24\sin\theta\cos\theta + 29\cos^2\theta)}$ is ' a ', then the value of a is ____.
17. Let $a = \sum_{r=1}^{11} \tan^2\left(\frac{r\pi}{24}\right)$ and $b = \sum_{r=1}^{11} (-1)^{r-1} \tan^2\left(\frac{r\pi}{24}\right)$, then the value of $\frac{a}{b}$ is ____.
18. Distressed PVR owners will open theater for public from the month of December. However, the government mandates that strict norms of social distancing are observed and no two people can take adjacent seats. Further erring on the side of caution, the owners decide to seat four people in a row which contains 15 seats. Find the total number of ways in which four seats can be selected out of 15 such that no two selections are consecutive.

SPACE FOR ROUGH WORK